

Uniform wire density

RUDY, Rectangular Uniform wire Density, spreads each net's wirelength evenly across GCells under its bounding box

The idea

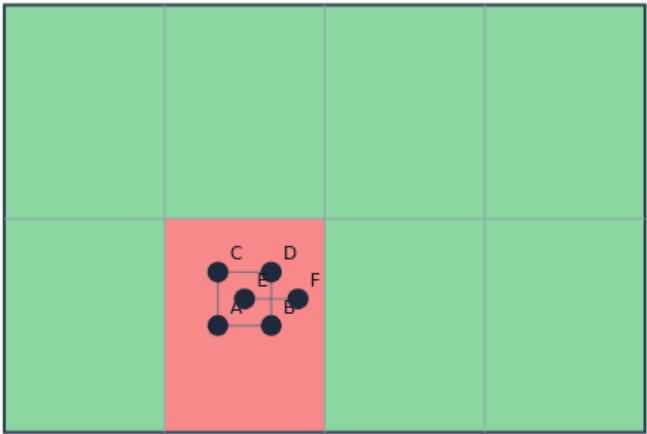
- For each net, take the axis-aligned bbox of pin positions
- Half-perimeter wirelength is width plus height
- Collect overlapping GCells, at least one
- Density equals HPWL divided by the tile count
- Add that density into every overlapping tile
- Sum across nets for the demand map

Net bounding box

RUDY ESTIMATE

Net bounding box

Step 1 / 5 · bbox · module02-01-rudy-estimate



Explanation

RUDY starts from each net's axis-aligned pin bbox and HPWL.

- $HPWL = width + height$

6 nets

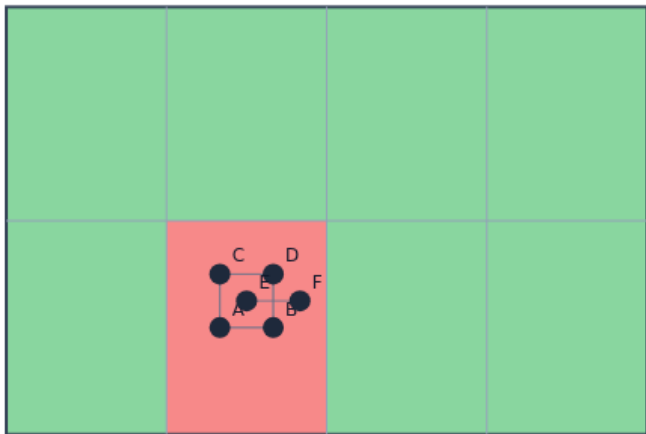
Net bounding box

Uniform share

RUDY ESTIMATE

Uniform share

Step 2 / 5 · share · module02-01-rudy-estimate



Explanation

Density = $\text{HPWL} / \text{\#overlapping GCells}$; add to each overlapping tile.

- At least one tile

Deterministic

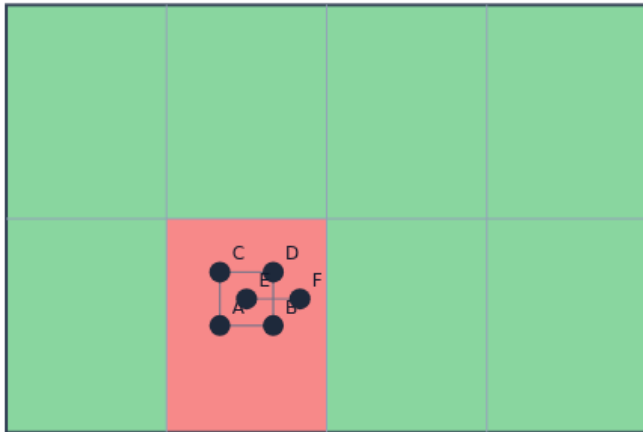
Uniform share

Sum over nets

RUDY ESTIMATE

Sum over nets

Step 3 / 5 · sum · module02-01-rudy-estimate



Explanation

Demand is the sum across nets—center tiles collect many contributions on a cluster.

- Matrix 4×2

Seed max ov ≈ 5

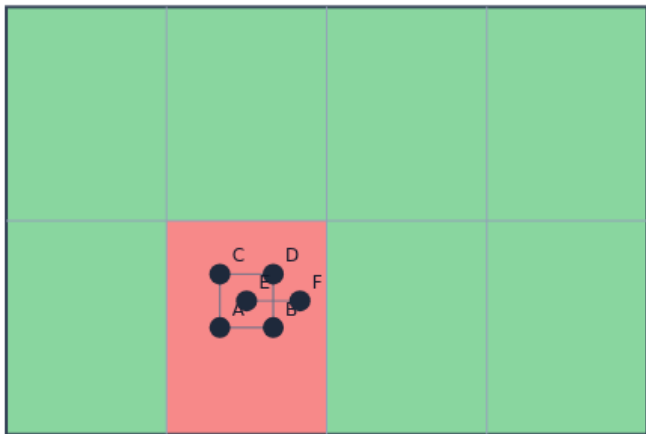
Sum over nets

Overflow appears

RUDY ESTIMATE

Overflow appears

Step 4 / 5 · overflow · module02-01-rudy-estimate



Explanation

$ov = \max(0, \text{demand} - \text{Cap})$. Seed shows a clear hotspot.

- total/max/count

Cap=2

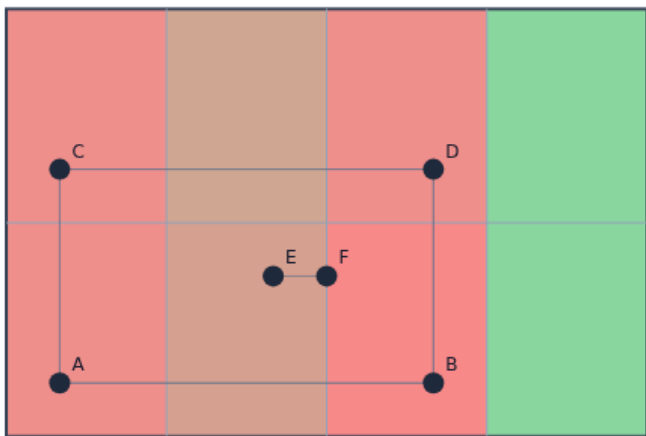
Overflow appears

Spread cools total pattern

RUDY ESTIMATE

Spread cools total pattern

Step 5 / 5 · spread · module02-01-rudy-estimate



Explanation

Long nets paint many tiles; cluster spikes max. Both are useful views.

- Compare seeds

Use max for hotspots

Spread cools total pattern

Browser lab track

EDA Algorithms Platform

HomeTools

Home / Tools / RUDY congestion estimate

INTERACTIVE TOOL

RUDY congestion estimate

Estimate routing demand with RUDY on a tiny GCell grid.

Your job: RUDY deposits HPWL density into GCells under each net bbox.
Spread cells to cut overflow. Seed max overflow ≈ 5 .

Reset workspace

Challenges0 / 8 cleared

Pick one [Intro] Seed max ≥ 4

Pick a challenge and click **Start**. Move cells, then **Check**. Reveal golden is study-only.

Start

Show hint

Check

Next challenge

Stop checking

Idle

A

B

C

D

E

F

←

→

↑

↓

Reset to starter

Reveal golden (study)

Run feedback

Graph

Reference / metrics

selected: A @ GCell (1,0)
HPWL: 7.0
overflow total: 5.00 · max: 5.00 · count: 1
capacity: 2
view: your placement

Browser lab starter

learn_congestion — 01 rudy estimate

Implement track

- Implement ``rudy_demand(positions)`` in ``common/solvers.py``
- On ``congested_seed``
- Match the browser golden within a small rounding tolerance

Pitfalls

- Dividing by bbox area in continuous units while depositing into discrete tiles
- Skipping nets with coincident pins, still touch one GCell
- Mutating the demand matrix in place across calls without zeroing

Your turn

- Ship Track A RUDY and clear the browser challenges
- Next: probabilistic L-shapes for a different demand signature